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① (a) Make the common Case Fast : Most of the programs spend most of their time executing a small portion of the code. If we optimize the part that execute frequently, overall performance improves significently.

(b) performance via parallelism: Instead of executing one task at a time, multiple tasks are executed simulteanously. parallelism increases throughput and can reduce execution time if the program is parallelizable.

(ii) Process of Building a processor

(i) Define ISA
(ii) Micro arch design
(iii) logic design
(iv) physical design
(v) Fabrication
(vi) testing & packing.

CamScanner

② (A) Best performance

passenger capacity : Airbus A380

speed : BAC

Range : Boeing 777

Yes, the perspective is very necessary because performance must always specify a metric. Therefore performance depends on perspective.

(B) Multicore processors definitely increase throughput by executing multiple independent task / threads simultaneously. For parallel programs → can reduce execution time. therefore multicore mainly increase throughput.

(3) Yes, Compiler behavior should be considered when calculating average cpI. Because compiler affects instruction mix. So that it's affects weighted average cpI.

Hardware based cpi: Based on actual instruction class cycle and fixed for the processor.

Software based cpi: Depends on instruction distribution of program.

So, that best is use software-based weighted average cpi because real program have mixed instruction.

(5) Mips is not a reliable pentormance metric because

(i) Different processors use different sets

(ii) CPI varies between programs

(iii) instruction vary in complexity.

(iv) A higher MIPS dose not necessarily mean lower execution time.

execution time is a more accurate pentormance metric.

$$\text{New time} = \frac{\text{Time affected}}{\text{Improvement}} + \text{Time unaffected}$$

This law implies that improving a small portion of a program yields limited over all improvement.

* if 80 seconds out of 100 sec. are spent on multiplication and we improve multiplication by 2x

$$\text{New Time} = \frac{80}{2} + 20 = 60$$

Speedup = 100/60 = 1.67x

Even though multiplication was improved 2x, overall speed only 1.67x. this demonstrates why we must optimize the common case.

④ Improvement of processors over generation performance improved by increasing clock frequency

$$\text{cpu time} = \frac{I_c \times CPI}{\text{clock Rate}}$$

increasing clock rate reduces execution time

It' didn't sustain. Brew power = $ex \times f$

increasing frequency increases power and heat

Voltage scaling also stopped working due to physical limits. they come up with multiple parallel pipeline improvement

The final nail in the coff in is the power wall - we cannot increase clock rate further due to heat dissipation limits.